Losartan PBPK/PD model: fit PK

1 Methods: Losartan PBPK/PD model: fit PK

1.1 Parameters and data

14 parameters were optimized: tissue distribution LOS (`ftissue_los`), tissue/blood partition coefficient LOS (`Kp_los`), tissue distribution E3174 (`ftissue_e3174`), tissue/blood partition coefficient E3174 (`Kp_e3174`), rate of losartan absorption (`GU__LOSABS_k`), rate losartan → E3174 (`LI__LOS2E3174_k`), rate losartan → other metabolites (`LI__LOS2L158_k`), rate E3174 export (`LI__E3174EX_k`), rate E3174 import (`LI__E3174IM_k`), rate E3174 bile export (`LI__E3174BIEX_k`), rate bile export of the other metabolites (`LI__L158BIEX_k`), renal clearance of unbound losartan relative to the GFR (`KI__LOSEX_f`), renal clearance of unbound E3174 relative to the GFR (`KI__E3174EX_f`) and renal clearance of the other metabolites relative to the GFR (`KI__L158EX_f`).

A fit mapping is a curve of a study, the data and the observable of the model it is compared with. The fit has 72 fit mappings of training data with 586 data points, whose residuals enter the objective; and 117 fit mappings of validation data with 941 data points, which were simulated and evaluated with the fitted parameters but not fitted. The fit was set up and run with sbmlsim [1].

1.2 Objective function

The parameters were estimated by minimizing the objective function

$$F(\vec{p}) = \frac{1}{2} \sum_k \sum_i \rho\big(w_{i,k} r_{i,k}(\vec{p})^2\big) \tag{1}$$

where k runs over the fit mappings of the training data and i over their data points, $r_{i,k}$ is the residual, $w_{i,k} = w_k w^p_{i,k}$ the weight of the data point and ρ the loss function, applied to the squared weighted residual $z = w_{i,k} r^2_{i,k}$. The loss function is linear, $\rho(z) = z$, i.e. a standard least squares problem.

The residuals are normalized by the mean $\overline{y}_k$ of the data of their curve, which makes curves of different quantities and magnitudes comparable:

$$r_{i,k} = \frac{y_{i,k} - f_{k(x_{i,k},\vec{p})}}{\overline{y}_k} \tag{2}$$

The weight of curve k is $w_k = \frac{n_k}{N_k}$, the weight n_k of its fit mapping (the weight its fit mapping collection assigns, 1 by default, or, for a collection which uses the weights of its mappings, the weight of the mapping, by default the number of subjects of its data) divided by the number N_k of its data points.

A data point is weighted with the inverse of its coefficient of variation, $w^p_{i,k} = |y_{i,k} \frac{|}{\sigma_{i,k}}$, with $\sigma_{i,k}$ the standard deviation of the data point, or its standard error for data without a standard deviation. An error of zero is missing, a missing error of a curve with errors is replaced by the largest error of the curve, and the data points of a curve without errors have the weight $w^p_{i,k} = 2$, the weight of a coefficient of variation of 0.5.

1.3 Optimization

The optimizer searched the parameters in log10 space within their bounds; bounds, values and intervals are given in the units of the fit parameters, the units of the table of the parameters.

The objective was minimized in 3 runs with differential evolution [2] of SciPy [3] within the bounds of the parameters, with the strategy `best1bin`, a population of 15 times the number of parameters, at most 1000 generations, the relative tolerance 0.01 of convergence, the recombination constant 0.7 and a mutation dithered between 0.5 and 1. The initial population of a run was drawn by latin hypercube sampling [4], with a seed of every run derived from the seed 1234 of the fit. The best member of the final population was polished with the L-BFGS-B method of SciPy.

The convergence of the runs was assessed from the traces of their cost and the waterfall plot of their final costs, and the parameter set with the lowest cost was used.

The models were simulated with libRoadRunner [5] and its integrator CVODE [6] with the relative tolerance 10^{-6} and the absolute tolerances 10^{-10} of an amount, 10^{-10} of a concentration and 10^{-6} of the other states, which for a species in a concentration is multiplied by the initial volume of its compartment, and 4 states of its own. It ran with the variable step size of libRoadRunner, i.e. with an output at every step of the integrator. A data point at the time of a change after the start of a simulation, e.g. a dose or the reset of a collection, was compared with the state before the

change, except the first data point of a curve with data at later times, which starts a collection and was compared with the state after the change.

1.4 Goodness of fit

The goodness of fit is given per kind of data (training data and validation data) by metrics over its n data points with the residuals $e_i = y_i - f\left(x_i, \hat{\vec{p}}\right)$ at the fitted parameters $\hat{\vec{p}}$, the mean squared error, its root and the root mean squared error normalized by the mean $\overline{y}_{k(i)}$ of the data of the curve of a point:

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n} e_i^2, \quad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}, \quad \mathrm{NRMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(\frac{e_i}{\overline{y}_{k(i)}}\right)^2} \tag{3}$$

The root mean square of the residuals of the objective, i.e. what the fit minimizes, and the coefficient of determination:

$$\mathrm{RMSE}_w = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\rho(w_i r_i^2)}, \quad R^2 = 1 - \frac{\sum_i e_i^2}{\sum_i \left(y_i - \overline{y}\right)^2} \tag{4}$$

The predictions of a non-linear model are no linear regression of the data, so R^2 can be negative: the prediction is worse than the mean of the data. The information criteria of Akaike [7] and Schwarz [8] with the number n_p of fitted parameters are those of a least squares fit with normally distributed residuals, up to an additive constant, so only differences between models fitted to the same data are meaningful:

$$\mathrm{AIC} = n\ln(\mathrm{MSE}) + 2n_p, \quad \mathrm{BIC} = n\ln(\mathrm{MSE}) + n_p\ln(n) \tag{5}$$

The percentage of the data points whose prediction $f_i = f\left(x_i, \hat{\vec{p}}\right)$ lies within twofold of the data is taken over the n_+ data points with positive data $y_i > 0$ and a positive prediction $f_i > 0$; the others have no ratio and are not counted, and a kind of data without such a point has no percentage:

$$P_2 = \frac{100}{n_+}\left|\left\{i : \frac{1}{2} \le \frac{f_i}{y_i} \le 2\right\}\right| \tag{6}$$

The goodness of fit plot shows the predictions against the data on logarithmic axes, with the lines of twofold $f = 2y$ and $f = \frac{y}{2}$. The Bland-Altman plot [9] shows the ratio of prediction and data as $d_i = \log_{10}\left(\frac{f_i}{y_i}\right)$ against the geometric mean $\sqrt{f_i y_i}$ for the data points with positive values, with the bias $\overline{d}$ and the limits of agreement $\overline{d} \pm 1.96 s_d$, with the sample standard deviation s_d of d_i, of the training data; they are read as the fold factors $10^{\overline{d}}$ and $10^{\overline{d} \pm 1.96 s_d}$.

1.5 Identifiability

The identifiability of the parameters was analysed with the profile likelihood [10], [11]. The profile of a parameter is the lowest objective with the parameter fixed:

$$\mathrm{PL}(p_j) = \min_{p_{l \ne j}} F(\vec{p}) \tag{7}$$

At every point of a scan the other parameters were re-optimized from the previous point with the least squares optimizer of SciPy with the relative step 0.05. A scan ran from the optimum in both directions in the space of the optimizer, with steps in decades of the parameter, starting at 0.1. A step which raised the objective by more than 0.2 of the distance between the objective at the optimum and the threshold was reduced by the factor 2 and repeated, down to 0.01, and the step after one which raised it by less than that divided by the factor was enlarged by the factor, up to 1. A scan stopped at the threshold, at the bound of the parameter, at a point which failed or after 50 points.

The confidence intervals are the values at which the profile crosses the threshold of the likelihood ratio test

$$F_{\mathrm{thr}} = F_{\mathrm{min}} + \chi^2_\alpha \frac{\mathrm{df}}{2} \tag{8}$$

with F_{min} the lowest objective of the fit and of all profiles and $\chi^2_\alpha(\mathrm{df})$ the quantile of the χ^2 distribution at the confidence level $\alpha = 0.95$ with df = 1 degrees of freedom, i.e. 1.92 above the lowest objective. The threshold has this confidence level if F is the negative log-likelihood of the data up to a constant, i.e. if the weighted residuals are independent and normally distributed with unit variance; otherwise the level is approximate. A crossing is interpolated between two points of the profile, on the logarithm of a parameter on a logarithmic scale. A parameter is identifiable if its profile reaches the threshold within the scan on both sides of the optimum, i.e. its confidence

interval is finite; practically non-identifiable towards the lower bound if its profile reaches the threshold above the optimum but not below it within the scan, which ends at the lower bound or after 50 points; practically non-identifiable towards the upper bound if its profile reaches the threshold below the optimum but not above it within the scan, which ends at the upper bound or after 50 points; practically non-identifiable if its profile reaches the threshold on neither side within the scans, which end at the bounds or after 50 points, and is not flat; and structurally non-identifiable if its profile is flat on both sides. A side of a profile is flat if it rises by less than 0.05 of the distance between the lowest objective and the threshold, and a profile with a point which failed or did not converge is not classified. The profile likelihood is preferred over the intervals of the Fisher information, its intervals are invariant under a transformation of the parameters and can be asymmetric [12].

The Fisher information is the local alternative to the profiles: it is the Gauss-Newton approximation of the curvature of the objective at the optimum, from the Jacobian J of the weighted residuals of the training data with respect to the parameters $\vec{p}$ in the space of the optimizer, computed by central finite differences with the step $h_j = \varepsilon \max(|p_j|, 1),\ \varepsilon = 10^{-6}$ of the parameter p_j:

$$I = J^\top J, \quad \mathrm{cov} = \sigma^2 I^{-1}, \quad \sigma^2 = \frac{2F}{n - n_p} \tag{9}$$

with the number n of data points of the training data and the number n_p of parameters. The information has rank 14 of 14, the number of its eigenvalues above 10^{-8} of the largest one, and the condition number $2.95 \cdot 10^{6}$; the covariance of an information of a lower rank is computed with the pseudo-inverse. The Wald intervals are $\hat{p}_j \pm t_{\frac{1+\alpha}{2}, n-n_p} \mathrm{SE}_j$ with the standard errors $\mathrm{SE}_j = \sqrt{\mathrm{cov}_{jj}}$ at the confidence level $\alpha = 0.95$, transformed back into the units of the fit parameters.

1.6 Provenance

The fit PK_20261019_113633__d9465 was run on 2026-10-19 09:36:38 UTC with sbmlsim 0.10.0 [1], libRoadRunner 2.10.0 [5], SciPy 1.18.1 [3], NumPy 2.5.3 [13] and Python 3.14.4 (seed 1234). The models and data were those of commit `8a4ec8bb1deeff002869fec7dcbd271c96a53d24` of `/home/mkoenig/git/pkdb_models-losartan` their SHA-256 checksums are listed in Table 1.

Table 1: SHA-256 checksums of the models and data of the fit PK_20261019_113633__d9465, their first 12 characters; the full checksums are in the provenance of the result (optimization_result.json).

File	SHA-256
pkdb_models/models/losartan/data/caffeine/Oh2012/.Oh2012_Fig3.tsv	e60ef89928e7...
pkdb_models/models/losartan/data/caffeine/Oh2012/.Oh2012_TabGroups.tsv	021f55009aad...
pkdb_models/models/losartan/data/caffeine/Oh2012/.Oh2012_TabInterventions.tsv	2ccc461e26f0...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_Fig3A.tsv	d3c6049b450f...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_Fig3B.tsv	86b2d598853f...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_Fig3C.tsv	b615f51b6bb6...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_Fig3D.tsv	035aba0ccaec...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_Fig3E.tsv	b2c71cf1bb7c...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_Tab5.tsv	aa23acadd634...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_TabGroups.tsv	2d6d7efcfbec...
pkdb_models/models/losartan/data/caffeine/Tanaka2014/.Tanaka2014_TabInterventions.tsv	ddded3dea742...
pkdb_models/models/losartan/data/chlorzoxazone/Puris2019/.Puris2019_Fig1.tsv	655030c953c5...
pkdb_models/models/losartan/data/chlorzoxazone/Puris2019/.Puris2019_Tab2.tsv	118f0088ed72...
pkdb_models/models/losartan/data/chlorzoxazone/Puris2019/.Puris2019_TabGroups.tsv	e515a75bff14...
pkdb_models/models/losartan/data/chlorzoxazone/Puris2019/.Puris2019_TabInterventions.tsv	d447ba94154e...
pkdb_models/models/losartan/data/chlorzoxazone/Puris2019/.Puris2019_TabS3.tsv	1bfb438995b3...
pkdb_models/models/losartan/data/chlorzoxazone/Puris2019/.Puris2019_TabS4.tsv	61f03a6b5f6d...

File	SHA-256
pkdb_models/models/losartan/data/dextromethorphan/Kim2016/.Kim2016_Fig1.tsv	c44528ec99d6...
pkdb_models/models/losartan/data/dextromethorphan/Kim2016/.Kim2016_Tab2.tsv	ebf3972b7a8d...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Fig1.tsv	6a94b463a1a9...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Fig1A.tsv	9fccde15cbcf...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Fig2.tsv	09b76010486d...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Fig3.tsv	15959fdca948...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Fig4.tsv	953290847694...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Fig5.tsv	e3ac24952f91...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Tab1.tsv	67e8ad2df53b...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_Tab2.tsv	0f7eb7fc7843...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_TabGroups.tsv	d8ad24602406...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_TabIndividuals.tsv	942da8c1d6f0...
pkdb_models/models/losartan/data/losartan/Azizi1999/.Azizi1999_TabInterventions.tsv	07e29032bfad...
pkdb_models/models/losartan/data/losartan/Bae2011/.Bae2011_Fig1.tsv	c064ae630cc4...
pkdb_models/models/losartan/data/losartan/Bae2011/.Bae2011_Tab4.tsv	1be68f5a5d24...
pkdb_models/models/losartan/data/losartan/Bae2011/.Bae2011_TabGroups.tsv	7f3c372a49d9...
pkdb_models/models/losartan/data/losartan/Bae2011/.Bae2011_TabInterventions.tsv	7834782f5928...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig1.tsv	a59ac80ea2c2...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig2.tsv	40c8679dde83...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig2A.tsv	40b594aa4902...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig3.tsv	8a059dea6437...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig4.tsv	c89c02ea95d7...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig5.tsv	23c3e4ca55ac...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig6.tsv	74606074cc22...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Fig7.tsv	619954aecce8...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Tab1.tsv	cf8c1f921eae...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_Tab2.tsv	17a325111a09...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_TabGroups.tsv	6e44cdbdc84d...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_TabIndividuals.tsv	a5359065a28f...
pkdb_models/models/losartan/data/losartan/Christen1991a/.Christen1991a_TabInterventions.tsv	3e9da7607f61...
pkdb_models/models/losartan/data/losartan/Doig1993/.Doig1993_Fig1.tsv	4042469aae50...
pkdb_models/models/losartan/data/losartan/Doig1993/.Doig1993_Fig2.tsv	9f453a3ef799...
pkdb_models/models/losartan/data/losartan/Doig1993/.Doig1993_Tab4.tsv	8fd1cc52c3a5...
pkdb_models/models/losartan/data/losartan/Doig1993/.Doig1993_TabGroups.tsv	d1f5f840fd81...
pkdb_models/models/losartan/data/losartan/Doig1993/.Doig1993_TabInterventions.tsv	665bebb83ead...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig1.tsv	0c25c2d2ea51...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig1A.tsv	152e7814ae3a...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig1B.tsv	e0b6940affd1...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig1C.tsv	e5cd078748a6...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig1D.tsv	6a524077d90c...

File	SHA-256
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig1E.tsv	8921290ac889...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig1F.tsv	7753a7a91fb5...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig2.tsv	d8a2233fb05a...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_Fig3.tsv	f6ed75bf7891...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_TabGroups.tsv	6155fc803d61...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_TabIndividuals.tsv	fadbdce5455a...
pkdb_models/models/losartan/data/losartan/Donzelli2014/.Donzelli2014_TabInterventions.tsv	a44e6da6107e...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_Fig1.tsv	c21fa402fe13...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_Tab60_1.tsv	6370229df392...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_Tab60_2.tsv	7261317e664e...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_Tab60_2A.tsv	84bc34a7f5e6...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_Tab60_2B.tsv	52e11b11cd83...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_TabGroups.tsv	d938463cefba...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_TabIndividuals.tsv	9ec28c2aad7f...
pkdb_models/models/losartan/data/losartan/FDA1995S60/.FDA1995S60_TabInterventions.tsv	7e65e3994c5a...
pkdb_models/models/losartan/data/losartan/FDA1995S67/.FDA1995S67_Fig1.tsv	99c530ccf784...
pkdb_models/models/losartan/data/losartan/FDA1995S67/.FDA1995S67_Tab1.tsv	2a117a95e309...
pkdb_models/models/losartan/data/losartan/FDA1995S67/.FDA1995S67_Tab1B.tsv	017964273f06...
pkdb_models/models/losartan/data/losartan/FDA1995S67/.FDA1995S67_TabGroups.tsv	1d4cb9ecd79c...
pkdb_models/models/losartan/data/losartan/FDA1995S67/.FDA1995S67_TabIndividuals.tsv	31e18bcc45fe...
pkdb_models/models/losartan/data/losartan/FDA1995S67/.FDA1995S67_TabInterventions.tsv	348bb267e1a9...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_Fig1.tsv	8bd34688a7a3...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_Fig2.tsv	0f93307f2ea1...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_Fig3.tsv	4549d5c65670...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_Tab1.tsv	bd520f015732...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_Tab2.tsv	91aeaec34cb8...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_Tab2A.tsv	04a60bf345e6...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_TabGroups.tsv	09ad283d841e...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_TabIndividuals.tsv	d4419bf54d77...
pkdb_models/models/losartan/data/losartan/Fischer2002/.Fischer2002_TabInterventions.tsv	b54215b38ad6...
pkdb_models/models/losartan/data/losartan/Goldberg1995/.Goldberg1995_Fig1.tsv	8c07d6f26cca...
pkdb_models/models/losartan/data/losartan/Goldberg1995/.Goldberg1995_Fig2.tsv	c6b0d0d9a82a...
pkdb_models/models/losartan/data/losartan/Goldberg1995/.Goldberg1995_Tab1.tsv	e18e6f9471d5...
pkdb_models/models/losartan/data/losartan/Goldberg1995/.Goldberg1995_TabGroups.tsv	7de68e35df2f...
pkdb_models/models/losartan/data/losartan/Goldberg1995/.Goldberg1995_TabInterventions.tsv	77000078bdf3...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Fig2.tsv	2070a3347963...

File	SHA-256
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Fig3.tsv	36138e6628e7...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Fig4.tsv	8535d3752708...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Fig5.tsv	df668ef0a0d4...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Fig6.tsv	3aa62b27763b...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Fig7.tsv	945c5840f387...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Tab1.tsv	2c0f00193067...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Tab2.tsv	c3399f096831...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_Tab3.tsv	58ab398b08de...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_TabGroups.tsv	058a98915c8d...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_TabIndividuals.tsv	3b4bf0f9c7c2...
pkdb_models/models/losartan/data/losartan/Goldberg1995a/.Goldberg1995a_TabInterventions.tsv	b3b0d326ffc9...
pkdb_models/models/losartan/data/losartan/Han2009a/.Han2009a_Fig1.tsv	fc31c3c12568...
pkdb_models/models/losartan/data/losartan/Han2009a/.Han2009a_Fig2.tsv	c8c387924e97...
pkdb_models/models/losartan/data/losartan/Han2009a/.Han2009a_Tab1.tsv	4e2d160252cf...
pkdb_models/models/losartan/data/losartan/Han2009a/.Han2009a_TabGroups.tsv	fb294e1561dd...
pkdb_models/models/losartan/data/losartan/Han2009a/.Han2009a_TabIndividuals.tsv	b3cbd106b395...
pkdb_models/models/losartan/data/losartan/Han2009a/.Han2009a_TabInterventions.tsv	f6510cba7ad2...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_Fig2.tsv	a9abc6e5b144...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_Fig3.tsv	d173fb346a9a...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_Fig4.tsv	745dec1a460e...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_Tab2.tsv	d53f3bef9a53...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_Tab3.tsv	ca41f2ce1dfc...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_TabGroups.tsv	44e079c8f3a1...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_TabIndividuals.tsv	c167ba44dd37...
pkdb_models/models/losartan/data/losartan/Huang2021/.Huang2021_TabInterventions.tsv	78612df49059...
pkdb_models/models/losartan/data/losartan/Kobayashi2008/.Kobayashi2008_Fig1.tsv	48f009fde5f8...
pkdb_models/models/losartan/data/losartan/Kobayashi2008/.Kobayashi2008_Tab1.tsv	841b99a46ccb...
pkdb_models/models/losartan/data/losartan/Kobayashi2008/.Kobayashi2008_TabGroups.tsv	7a927d53b3ab...
pkdb_models/models/losartan/data/losartan/Kobayashi2008/.Kobayashi2008_TabInterventions.tsv	95245c7bd501...
pkdb_models/models/losartan/data/losartan/Lee2003b/.Lee2003b_Fig1.tsv	d9868d7bed7d...
pkdb_models/models/losartan/data/losartan/Lee2003b/.Lee2003b_Tab1.tsv	068fea01a957...
pkdb_models/models/losartan/data/losartan/Lee2003b/.Lee2003b_TabGroups.tsv	855bd0dc6251...
pkdb_models/models/losartan/data/losartan/Lee2003b/.Lee2003b_TabInterventions.tsv	36f565cb3fae...
pkdb_models/models/losartan/data/losartan/Li2009/.Li2009_Fig1.tsv	501573f906be...
pkdb_models/models/losartan/data/losartan/Li2009/.Li2009_Tab2.tsv	cfdf04998d98...
pkdb_models/models/losartan/data/losartan/Li2009/.Li2009_TabGroups.tsv	360f966a359b...
pkdb_models/models/losartan/data/losartan/Li2009/.Li2009_TabInterventions.tsv	847db02a0e45...

File	SHA-256
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Fig1.tsv	2efbf4757755...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Fig2.tsv	dc34fc36e639...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab1.tsv	0627c2a4c8ae...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab1A.tsv	6c275f3ae2d0...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab1B.tsv	487dfe09dfd5...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab2.tsv	76820a6128cf...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab2A.tsv	433bb67d263c...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab2B.tsv	6f92d3a78662...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab3.tsv	9c990116c9b6...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab3A.tsv	fa15fe578097...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab3B.tsv	ad850d0cc936...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab4.tsv	1474e2e0a690...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_Tab4A.tsv	34c6c3c60d7a...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_TabGroups.tsv	a20603cb621f...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_TabIndividuals.tsv	aa7bde9a12bc...
pkdb_models/models/losartan/data/losartan/Lo1995/.Lo1995_TabInterventions.tsv	6e9d9fa12de5...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_Fig1.tsv	cb479ee413cd...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_Fig2.tsv	de1ad16348e0...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_Fig3.tsv	5bb39cedc6e6...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_Fig3A.tsv	18e6d4812024...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_Fig4.tsv	440f14855a04...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_Fig5.tsv	5cb0abcdfded...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_Fig6.tsv	4e0cd4817e37...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_TabGroups.tsv	334b899c5c18...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_TabIndividuals.tsv	618365c06a18...
pkdb_models/models/losartan/data/losartan/Munafo1992/.Munafo1992_TabInterventions.tsv	848741b84049...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Fig1.tsv	ba2286aef5e9...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Fig2.tsv	fa3b3a1f9c42...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Fig3.tsv	8e1f6c68b4a1...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Fig4.tsv	59bfc02a64e1...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Tab1.tsv	80b9a05004b7...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Tab2.tsv	0d08d74e8c9d...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Tab2A.tsv	477804665d07...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Tab3.tsv	7787f1d6079b...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_Tab4.tsv	273635c02d6e...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_TabGroups.tsv	63f27119bf00...
pkdb_models/models/losartan/data/losartan/Ohtawa1993/.Ohtawa1993_TabInterventions.tsv	143b0732e772...
pkdb_models/models/losartan/data/losartan/Sekino2003/.Sekino2003_Fig1.tsv	7abf4402adec...
pkdb_models/models/losartan/data/losartan/Sekino2003/.Sekino2003_Tab1.tsv	699c9f0fb3a4...
pkdb_models/models/losartan/data/losartan/Sekino2003/.Sekino2003_Tab1A.tsv	f381bc40fb37...
pkdb_models/models/losartan/data/losartan/Sekino2003/.Sekino2003_TabGroups.tsv	c3ee84511d8b...

File	SHA-256
pkdb_models/models/losartan/data/losartan/Sekino2003/.Sekino2003_TabInterventions.tsv	6e859af3177b...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_Fig1.tsv	ff0ae7c1e4b2...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_Fig2.tsv	c48800f37127...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_Tab2.tsv	7a4c2788ae12...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_Tab3.tsv	15e50c2f044c...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_Tab4.tsv	83cb1b07e887...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_TabGroups.tsv	456ca8d500c0...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_TabIndividuals.tsv	a180faf3e67f...
pkdb_models/models/losartan/data/losartan/Shin2020/.Shin2020_TabInterventions.tsv	4bcf9af32a88...
pkdb_models/models/losartan/data/losartan/Sica1995/.Sica1995_Fig1.tsv	36b10f7a5472...
pkdb_models/models/losartan/data/losartan/Sica1995/.Sica1995_Fig2.tsv	ebf8c317e078...
pkdb_models/models/losartan/data/losartan/Sica1995/.Sica1995_Fig3.tsv	0ccaa0f70727...
pkdb_models/models/losartan/data/losartan/Sica1995/.Sica1995_Fig4.tsv	7c4c6d82cbdc...
pkdb_models/models/losartan/data/losartan/Sica1995/.Sica1995_Tab2.tsv	87eacd3141b6...
pkdb_models/models/losartan/data/losartan/Sica1995/.Sica1995_TabGroups.tsv	137980523f53...
pkdb_models/models/losartan/data/losartan/Sica1995/.Sica1995_TabInterventions.tsv	f0268e49c99c...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_Fig1.tsv	853ab007c5cf...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_Fig2.tsv	081c8c68e4f1...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_Tab1.tsv	0274a8d09729...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_Tab2.tsv	8c80933cafcb...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_Tab2A.tsv	ef795b234916...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_Tab3.tsv	8ec143a94a8b...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_Tab4.tsv	e4ea62a8dd5f...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_TabGroups.tsv	c3e180feba2c...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_TabIndividuals.tsv	ddbc35360002...
pkdb_models/models/losartan/data/losartan/Yasar2002a/.Yasar2002a_TabInterventions.tsv	8c946bce36dd...
pkdb_models/models/losartan/models/output/models/losartan_body_flat.xml	c8d3686fe4d1...

2 Results: Losartan PBPK/PD model: fit PK

2.1 Parameters

Table 2: Parameters of the fit PK_20261019_113633__d9465, the parameter set PK_20261019_113633__d9465_0, cost 141, with their bounds in the optimization. The 95% confidence intervals are those of the profile likelihood, LB and UB an interval which is open towards the lower or the upper bound, and the Wald intervals of the Fisher information for a parameter without a profile. Identifiable is the classification of the profile: yes, no (LB) or no (UB) if the profile does not reach the threshold towards the lower or the upper bound, no if it reaches it on neither side, structural if it is flat.

Parameter	Description	Value [95% CI]	Unit	Bounds	Identifiable
ftissue_los	tissue distribution LOS	1.04 [0.447, 1.87]	l/min	[0.05, 5]	yes
Kp_los	tissue/blood partition coefficient LOS	1.17 [0.686, 2.17]	dimensionless	[0.1, 5]	yes
ftissue_e3174	tissue distribution E3174	0.00217 [LB, UB]	l/min	[0.001, 0.1]	no

Parameter	Description	Value [95% CI]	Unit	Bounds	Identifiable
Kp_e3174	tissue/blood partition coefficient E3174	0.186 [LB, UB]	dimensionless	[0.05, 2]	no
GU__LOSABS_k	rate of losartan absorption	0.0476 [0.0385, 0.0586]	1/min	[0.001, 0.5]	yes
LI__LOS2E3174_k	rate losartan → E3174	0.413 [0.223, 1.12]	1/min	[0.01, 10]	yes
LI__LOS2L158_k	rate losartan → other metabolites	1.79 [1.11, 2.16]	1/min	[0.1, 20]	yes
LI__E3174EX_k	rate E3174 export	0.00411 [0.00107, 0.0118]	1/min	[0.0001, 1]	yes
LI__E3174IM_k	rate E3174 import	0.0614 [0.0312, 0.135]	1/min	[0.001, 10]	yes
LI__E3174BIEX_k	rate E3174 bile export	0.00274 [0.00175, 0.00405]	1/min	[1e-05, 0.1]	yes
LI__L158BIEX_k	rate bile export of the other metabolites	0.257 [0.0421, 0.584]	1/min	[0.001, 1]	yes
KI__LOSEX_f	renal clearance of unbound losartan relative to the GFR	59.6 [52, 69.5]	dimensionless	[5, 500]	yes
KI__E3174EX_f	renal clearance of unbound E3174 relative to the GFR	125 [105, 142]	dimensionless	[10, 1e+03]	yes
KI__L158EX_f	renal clearance of the other metabolites relative to the GFR	5.19 [LB, UB]	dimensionless	[0.05, 10]	no

2.2 Data

Table 3: Data of the fit PK_20261019_113633__d9465: the number of fit mappings and, in parentheses, of data points per study and kind. The training data enters the cost, the validation data and the outliers are evaluated only.

Study	Training	Validation	Outlier	Total
Azizi1999	-	2 (14)	-	2 (14)
Bae2011	2 (19)	2 (22)	-	4 (41)
Christen1991a	2 (17)	-	-	2 (17)
Donzelli2014	-	2 (21)	-	2 (21)
FDA1995S60	6 (35)	5 (10)	-	11 (45)
FDA1995S67	-	10 (81)	-	10 (81)
Fischer2002	4 (21)	4 (22)	-	8 (43)
Goldberg1995	2 (37)	2 (37)	-	4 (74)
Goldberg1995a	4 (46)	-	-	4 (46)
Han2009a	2 (28)	6 (84)	-	8 (112)
Huang2021	2 (34)	6 (102)	-	8 (136)
Kim2016	-	4 (52)	-	4 (52)
Kobayashi2008	2 (16)	2 (16)	-	4 (32)
Lee2003b	2 (20)	4 (39)	-	6 (59)
Li2009	2 (25)	4 (50)	-	6 (75)
Lo1995	16 (120)	-	-	16 (120)
Munafo1992	6 (23)	-	-	6 (23)
Oh2012	-	2 (28)	-	2 (28)

Study	Training	Validation	Outlier	Total
Ohtawa1993	8 (80)	4 (58)	-	12 (138)
Puris2019	-	10 (28)	-	10 (28)
Sekino2003	2 (5)	2 (5)	-	4 (10)
Shin2020	4 (33)	8 (66)	-	12 (99)
Sica1995	-	12 (118)	-	12 (118)
Tanaka2014	-	4 (20)	-	4 (20)
Yasar2002a	6 (27)	22 (68)	-	28 (95)
Total	72 (586)	117 (941)	-	189 (1527)

2.3 Goodness of fit

Table 4: Goodness of fit of the fit PK_20261019_113633__d9465 per kind of data, the parameter set PK_20261019_113633__d9465_0. The cost is defined on the training data only, 2-fold is the percentage of the data points with a positive value and prediction whose prediction is within twofold of the data.

Kind	n	cost	MSE	RMSE	NRMSE	RMSE_w	R^2	2-fold [%]	AIC	BIC
training	586	141	0.327	0.571	0.542	0.694	0.742	76.5	−627.8	−566.5
validation	941	-	2.31	1.52	1.03	2.28	0.623	62.4	815.2	883.1

2.4 Figures

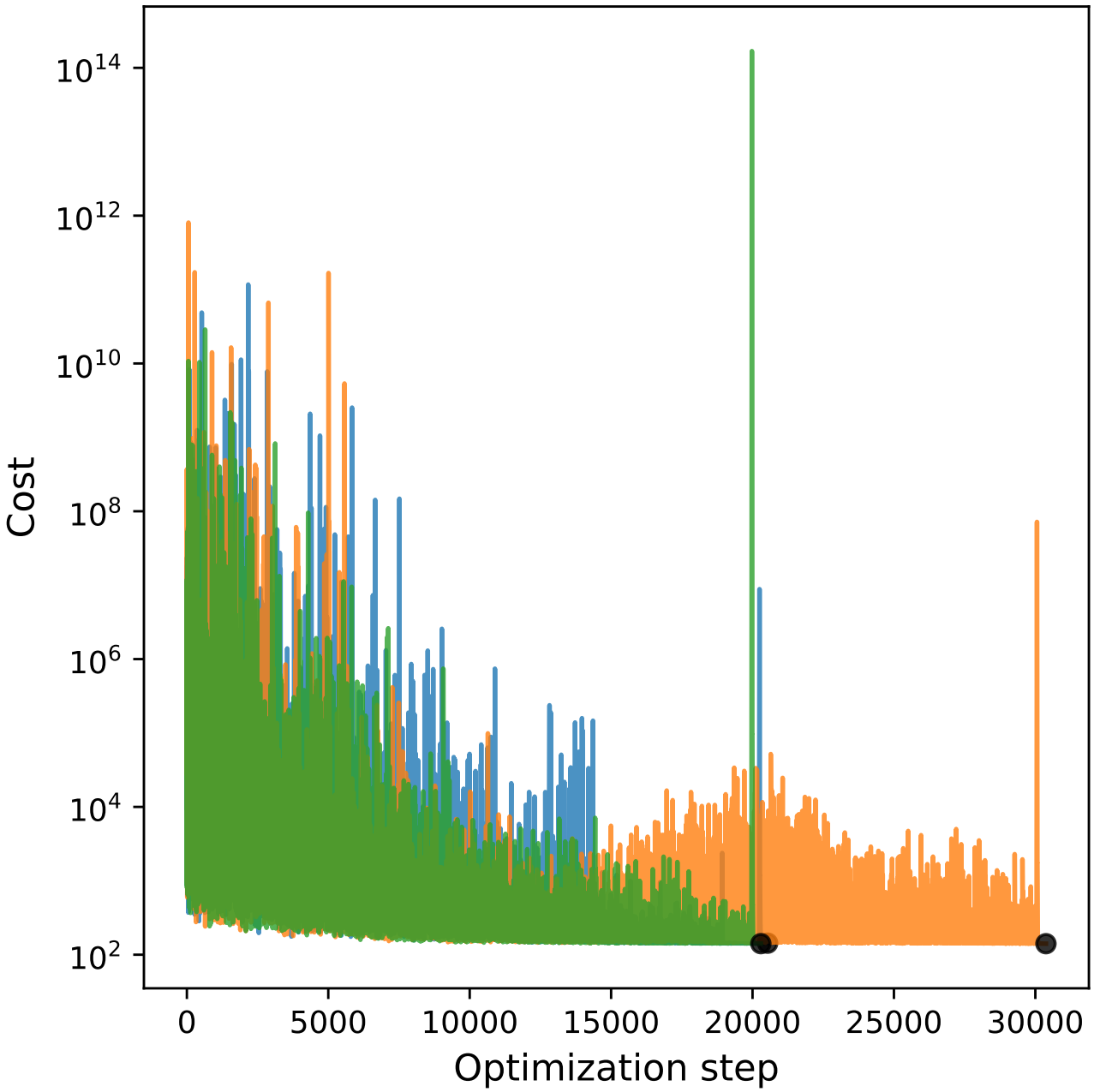


Figure 1: Cost of the optimizers over their steps, fit PK_20261019_113633__d9465. The cost of every optimization run over its steps. Runs which end at the same cost found the same optimum; a run which stops much higher is stuck in a local minimum.

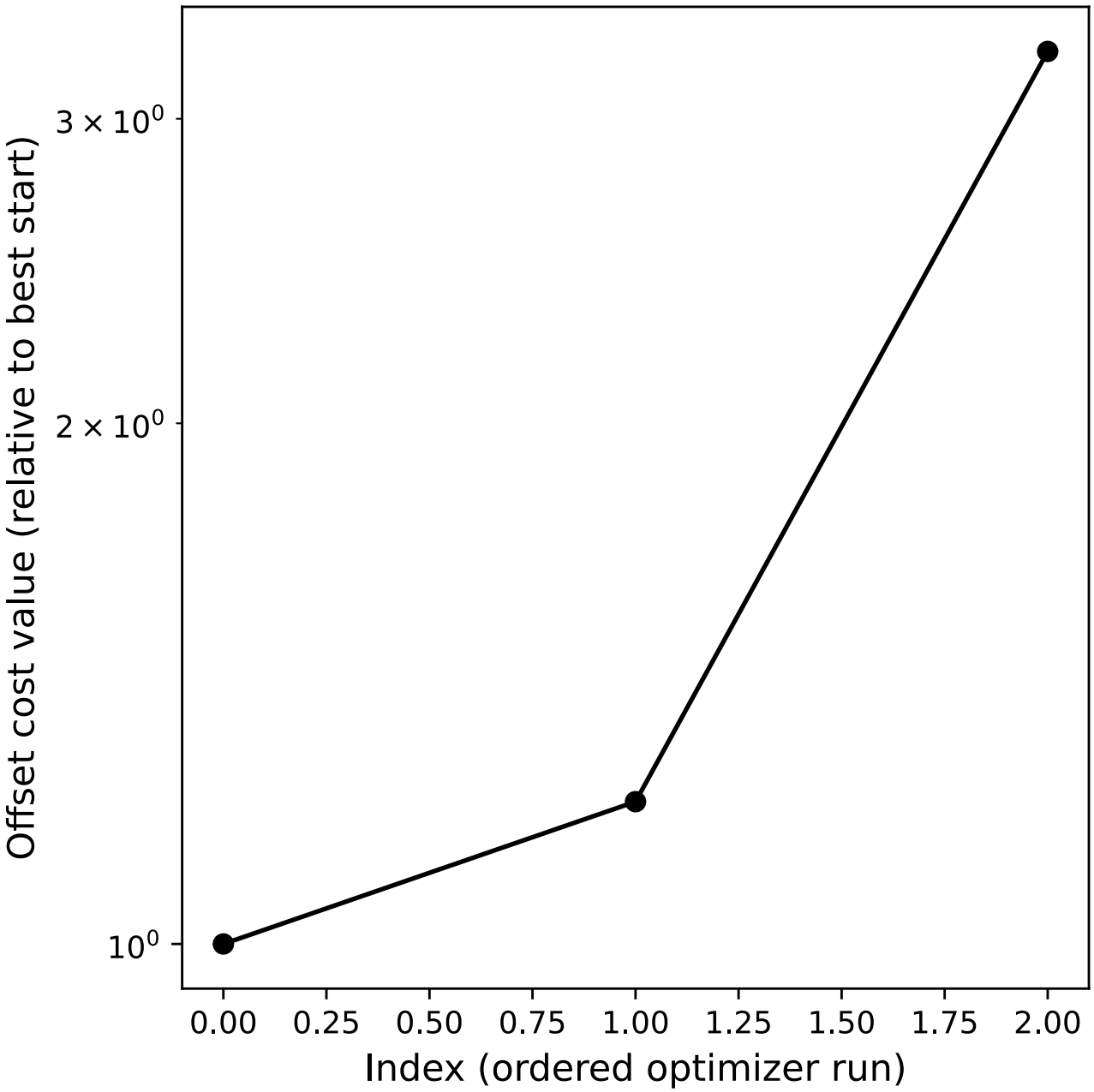


Figure 2: Cost of the optimization runs, ordered, fit PK_20261019_113633__d9465. The final cost of the runs, ordered. A flat plateau at the left is the global optimum found repeatedly, which is the evidence that the multistart converged; steps to the right are local minima.

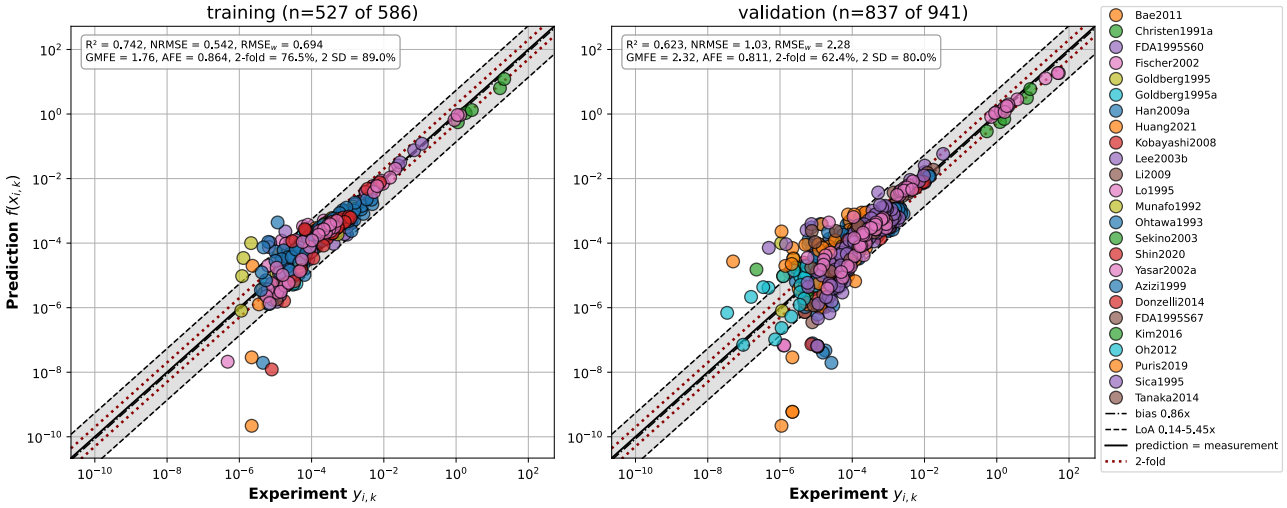


Figure 3: Prediction against the measured data points, per kind of fit mapping, with the agreement of the training data, the lines of twofold and the metrics of the panel, fit PK_20261019_113633__d9465. Within twofold of the data are 403 of the 527 training data points (76.5%) and 522 of the 837 validation data points (62.4%); only the data points with a positive value and prediction are counted. Prediction against measurement, one point per data point, with a panel per kind of fit mapping. The points scatter around the diagonal when the model describes the data; a systematic deviation from it is a systematic error of the model. The band is the agreement of the training data, i.e. the same band the Bland-Altman plot draws, here as lines parallel to the diagonal. The dark red dotted lines are twofold, prediction = 2 x measurement and prediction = measurement / 2; the caption states the points between them of every kind. A point which is zero or negative is not drawn on the logarithmic axes and not counted, the title of a panel then states the points drawn of all. The validation panel shows how the fit describes data it was not fitted on, the outlier panel the data the fit dropped. The box of a panel carries its R², its normalized RMSE, the RMSE of the residuals of the cost, its fold errors GMFE and AFE and the percentages of its points within twofold and within two standard deviations of the data.

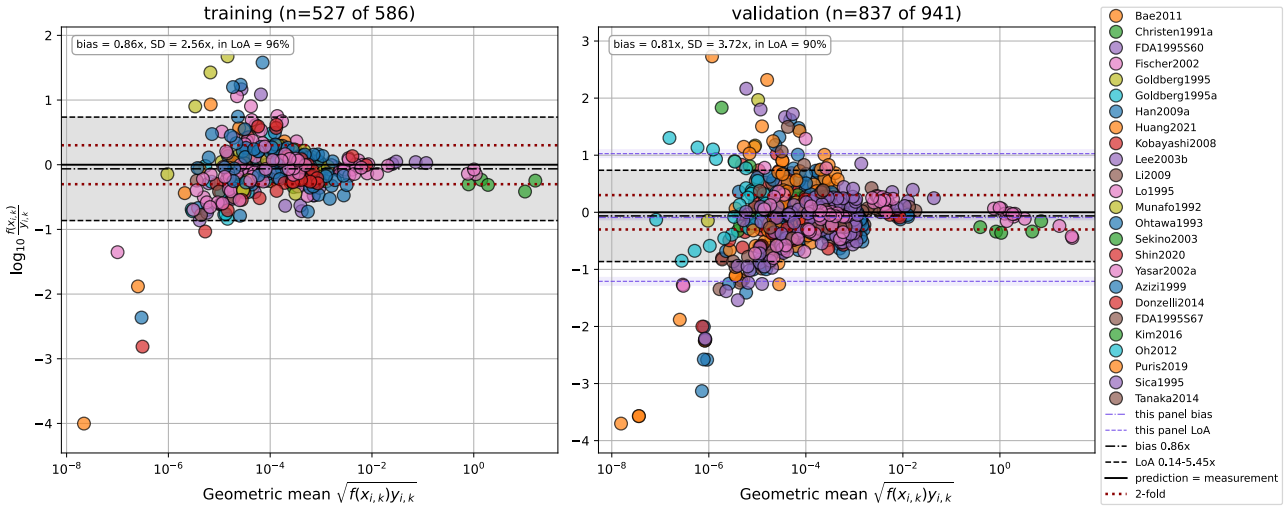


Figure 4: Agreement of prediction and measurement as a ratio, per kind of fit mapping, with the agreement of the training data and the agreement of the panel, fit PK_20261019_113633__d9465. The ratio of prediction and measurement over the geometric mean of the two, i.e. the goodness of fit with the diagonal turned into the horizontal. The band is the same, the bias and the limits of agreement `bias ± 1.96 SD` of the training data as fold factors. A bias away from 1 is a systematic over- or underprediction, wide limits are a large scatter, and a trend over the mean is a model which describes the large or the small values better. The box of a panel carries the bias and the SD of its own points as fold factors and the share of them inside the limits of agreement of the training data.

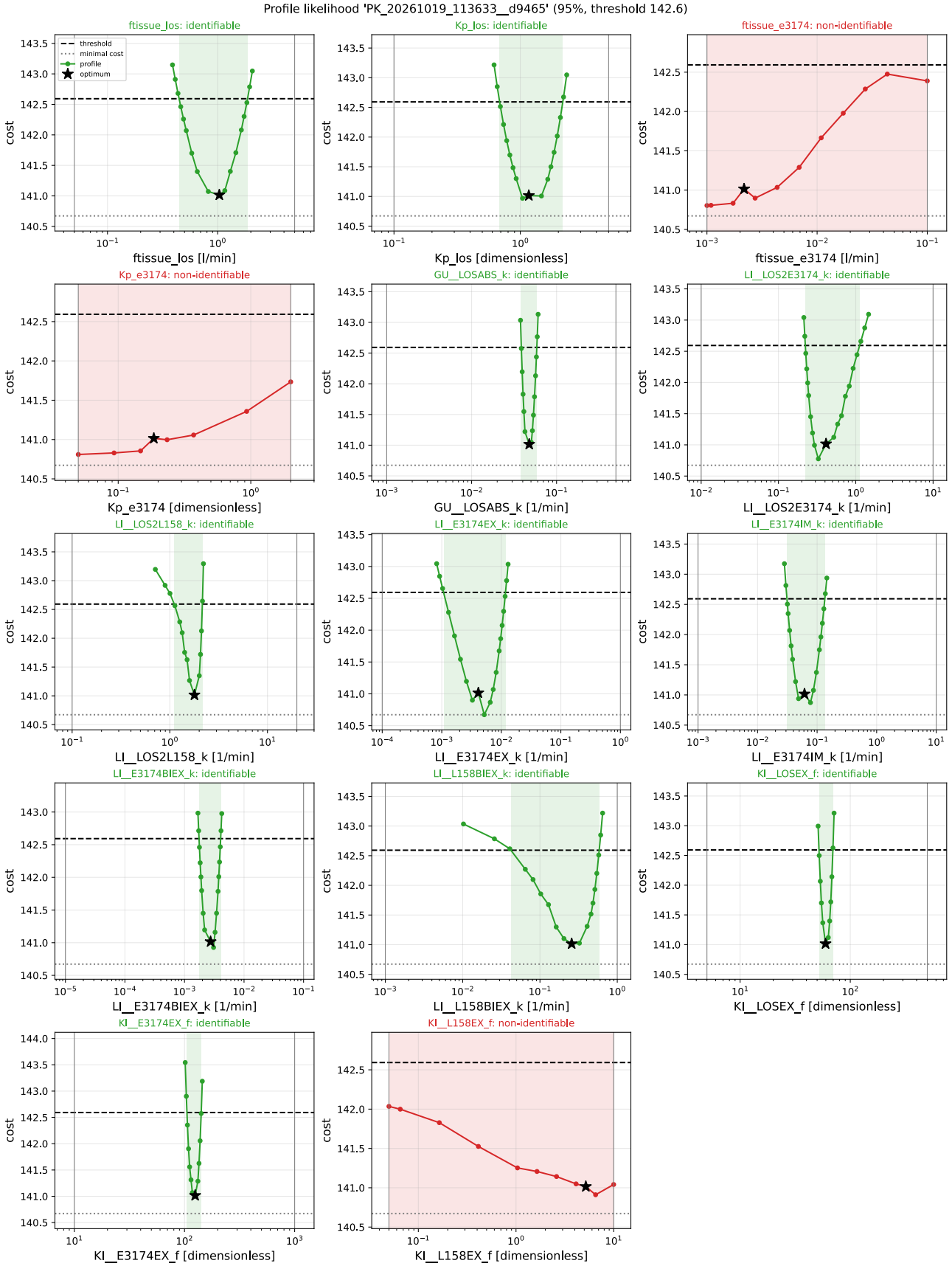


Figure 5: Profile likelihood of every parameter: the cost with the parameter fixed and the other parameters optimized, the threshold of the confidence level and the confidence interval, fit PK_20261019_113633__d9465. One profile per parameter. A profile which crosses the dashed threshold on both sides gives a finite confidence interval; one which stays below it up to a bound is practically non-identifiable; a flat profile is structurally non-identifiable, the parameter is compensated by the others.

2.5 Provenance

Table 5: Provenance of the fit PK_20261019_113633__d9465: the versions it ran with and the revision of its models and data; the SHA-256 checksums of the models and data are listed in Table 1.

Item	Value
Fit	`PK_20261019_113633__d9465`
Created	2026-10-19 09:36:38 UTC
Seed	1234
Python	3.14.4
sbmlsim	0.10.0
libRoadRunner	2.10.0
SciPy	1.18.1
NumPy	2.5.3
Repository	`/home/mkoenig/git/pkdb_models-losartan`
Commit	`8a4ec8bb1deeff002869fec7dcbd271c96a53d24`
Uncommitted changes	no

Bibliography

[1] M. König, "sbmlsim: SBML simulation made easy." Zenodo, 2026. doi: 10.5281/zenodo.3597770. Version 0.10.0.

[2] R. Storn and K. Price, "Differential evolution - a simple and efficient heuristic for global optimization over continuous spaces," *Journal of Global Optimization*, vol. 11, pp. 341–359, 1997, doi: 10.1023/A:1008202821328.

[3] P. Virtanen *et al.*, "SciPy 1.0: fundamental algorithms for scientific computing in Python," *Nature Methods*, vol. 17, pp. 261–272, 2020, doi: 10.1038/s41592-019-0686-2. Version 1.18.1.

[4] M. D. McKay, R. J. Beckman, and W. J. Conover, "A comparison of three methods for selecting values of input variables in the analysis of output from a computer code," *Technometrics*, vol. 21, no. 2, pp. 239–245, 1979, doi: 10.2307/1268522.

[5] C. Welsh, J. Xu, L. Smith, M. König, K. Choi, and H. M. Sauro, "libRoadRunner 2.0: a high performance SBML simulation and analysis library," *Bioinformatics*, vol. 39, no. 1, p. btac770, 2023, doi: 10.1093/bioinformatics/btac770. Version 2.10.0.

[6] A. C. Hindmarsh *et al.*, "SUNDIALS: suite of nonlinear and differential/algebraic equation solvers," *ACM Transactions on Mathematical Software*, vol. 31, no. 3, pp. 363–396, 2005, doi: 10.1145/1089014.1089020.

[7] H. Akaike, "A new look at the statistical model identification," *IEEE Transactions on Automatic Control*, vol. 19, no. 6, pp. 716–723, 1974, doi: 10.1109/TAC.1974.1100705.

[8] G. Schwarz, "Estimating the dimension of a model," *The Annals of Statistics*, vol. 6, no. 2, pp. 461–464, 1978, doi: 10.1214/aos/1176344136.

[9] J. M. Bland and D. G. Altman, "Statistical methods for assessing agreement between two methods of clinical measurement," *The Lancet*, vol. 327, no. 8476, pp. 307–310, 1986, doi: 10.1016/S0140-6736(86)90837-8.

[10] A. Raue *et al.*, "Structural and practical identifiability analysis of partially observed dynamical models by exploiting the profile likelihood," *Bioinformatics*, vol. 25, no. 15, pp. 1923–1929, 2009, doi: 10.1093/bioinformatics/btp358.

[11] C. Kreutz, A. Raue, D. Kaschek, and J. Timmer, "Profile likelihood in systems biology," *The FEBS Journal*, vol. 280, no. 11, pp. 2564–2571, 2013, doi: 10.1111/febs.12276.

[12] F.-G. Wieland, A. L. Hauber, M. Rosenblatt, C. Tönsing, and J. Timmer, "On structural and practical identifiability," *Current Opinion in Systems Biology*, vol. 25, pp. 60–69, 2021, doi: 10.1016/j.coisb.2021.03.005.

[13] C. R. Harris *et al.*, "Array programming with NumPy," *Nature*, vol. 585, pp. 357–362, 2020, doi: 10.1038/s41586-020-2649-2. Version 2.5.3.